



Eagle Exhumer - test report

EAGLE → KiCad 10 conversion with strong quality control: results on 18 real designs, four full corpus runs

8 October 2026 · plugin 0.9.7 · KiCad 10.0.6 · Z2Labs · github.com/z2labs/eagle-exhumer · GPL-3.0

Summary. 18 real EAGLE designs (11 Olimex open-hardware boards, 7 of Zoltan Doczi's PCB designs; 3,214 parts, 14,044 pads, 7,610 vias, 1-6 copper layers) were converted from scratch with Eagle Exhumer 0.9.7 on 8 October 2026. **All 18 pass the quality control, and the 6-fault self-check passed 6/6 on every design.** Compared with KiCad 10.0.6's native EAGLE import of the same files, open (split) schematic nets went from 49 to 0, lost EAGLE net names from 306 to 0, ERC errors from 367 to 16 and DRC shorting items from 347 to 26. This is the fourth full run of the corpus; the first one (7 October, morning) passed 14 of 18.

1. Scope and environment

All runs were made on one Windows desktop PC with KiCad 10.0.6 (stable), its bundled Python and kicad-cli. The board is imported with `kicad-cli pcb import --format eagle`, the schematic with KiCad's GUI EAGLE importer (there is no command-line schematic import), then the plugin's fixer and quality control run unattended in batch mode (`eagle2kicad_batch.py`). The native-import baseline was measured once on the full GUI import of the same files (run2) with `kicad-cli sch erc / pcb drc`; `kicad-cli` stops listing at 499 items per violation type, so DRC totals are lower bounds. Nothing in the corpus was hand-edited between import and measurement.

Design	Source	EAGLE	Cu layers	Parts	Pads	Vias
A10-A20-OLINUXINO-MICRO-4GB Rev C	Olimex	6.4	4	501	2278	1428
A10-A20-OLINUXINO-MICRO-4GB Rev D	Olimex	6.4	4	501	2278	1428
A20-OLINUXINO-MICRO 4GB REV E2	Olimex	6.4	4	501	2278	1428
A10-OLinuXino-Lime Rev A	Olimex	6.2	6	356	1754	953
AM3352-OLINUXINO Rev A-14122012	Olimex	6.2	4	405	1742	674
iMX233-OLINUXINO-MAXI Rev D	Olimex	6.5.0	4	191	778	357
iMX233-OLinuXino-Micro Rev D	Olimex	6.5.0	2	124	572	187
iMX233-OLinuXino-Micro Rev C	Olimex	6.4	2	124	571	183
LCD-OLINUXINO-7TS Rev A	Olimex	7.1.0	2	86	357	201
A13-LCD10TS Rev A	Olimex	6.4	2	91	293	99
RPism adapterboard 0V3	Z. Doczi	6.4	2	94	247	27
LCD-OLINUXINO-4.3TS Rev A	Olimex	7.1.0	2	38	239	156
SE880 module EFM32LG-STK3600	Z. Doczi	6.4	2	77	206	176
MAX9709 Mono Bridge AMP 1V2	Z. Doczi	6.4	2	19	117	105
RTL SDR DSDAMP 0V1	Z. Doczi	6.4	2	46	103	83
BDVLS 0V1	Z. Doczi	6.4	1	23	78	31
LS013B7DH03 TFT Module	Z. Doczi	6.4	2	20	78	67
RPi Speaker	Z. Doczi	6.4	2	17	75	27
Total (18 designs)				3214	14044	7610

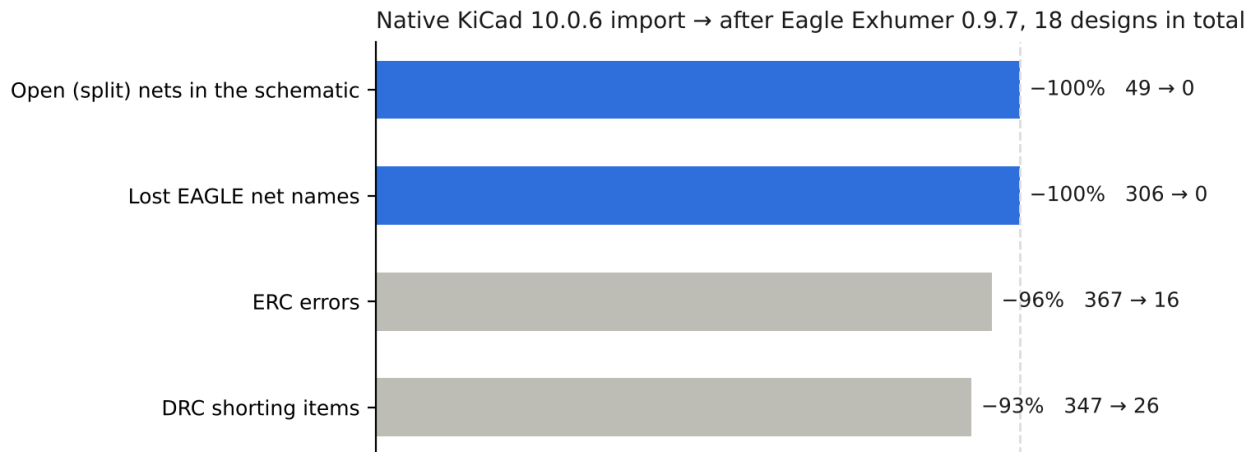
Olimex designs: github.com/OLIMEX/OLINUXINO (open hardware). A 19th, private 4-sheet EAGLE 9.6.2 design is part of the internal test set but is not reported here; it fails the quality control by design (a GND/GND0 reference collision that needs a human decision).

2. What is measured

Each design is converted twice for measurement purposes: once natively (KiCad's importer alone) and once through Eagle Exhumer. On both results the same four KiCad-native counts are taken: **open nets** in the schematic (one EAGLE net imported as several KiCad nets), **lost net names** (an EAGLE net name that no longer exists in the KiCad netlist), **ERC errors** and **DRC shorting items**. In addition, the plugin's own quality control compares the converted project with the original EAGLE files and returns a strict PASS/FAIL verdict, and a self-check injects six deliberate faults into a scratch copy of the result to prove that the quality control sees them. A design counts as passed only with

QC PASS and self-check 6/6.

3. Results, run4 (0.9.7)

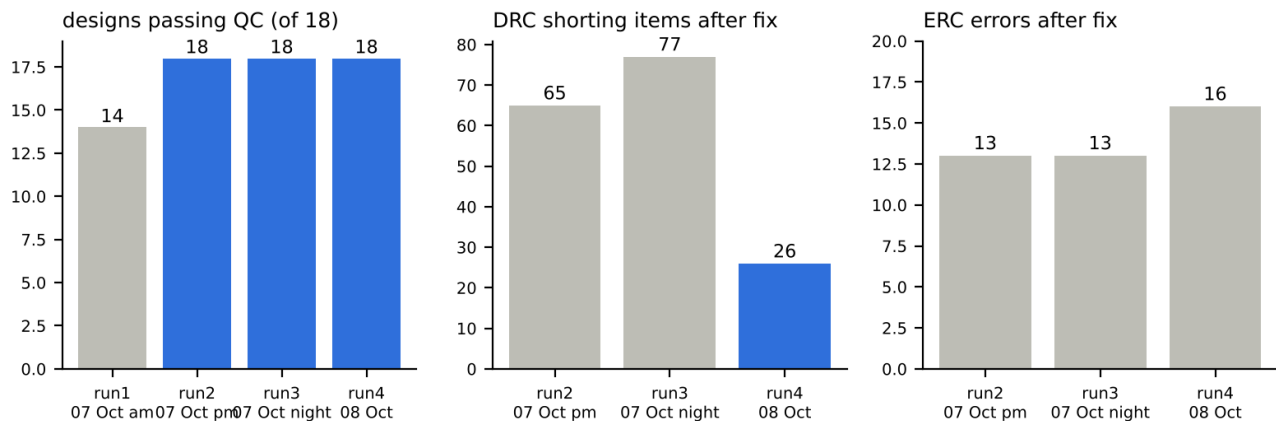


Per design. "Native" and "after" columns are open nets / lost net names / ERC errors / DRC shorting items. Times: board import (kicad-cli) + schematic import (GUI) = import; fix + quality control + renders = fix. Self-check 6/6 everywhere.

Design	Native: open / lost names	After 0.9.7: open / lost / ERC / short	Imports	Fixes	QC	Self-check
A10-A20-OLINUXINO-MICRO-4GB Rev C	16 / 80	0 / 0 / 2 / 0	29	200	PASS	6/6
A10-A20-OLINUXINO-MICRO-4GB Rev D	16 / 80	0 / 0 / 3 / 0	29	204	PASS	6/6
A20-OLINUXINO-MICRO 4GB REV E2	16 / 80	0 / 0 / 2 / 0	30	206	PASS	6/6
A10-OLinuXino-Lime Rev A	1 / 10	0 / 0 / 2 / 0	30	165	PASS	6/6
AM3352-OLINUXINO Rev A-14122012	0 / 21	0 / 0 / 2 / 1	27	121	PASS	6/6
iMX233-OLINUXINO-MAXI Rev D	0 / 3	0 / 0 / 2 / 7	27	93	PASS	6/6
iMX233-OLinuXino-Micro Rev D	0 / 3	0 / 0 / 1 / 4	30	77	PASS	6/6
iMX233-OLinuXino-Micro Rev C	0 / 3	0 / 0 / 1 / 4	26	73	PASS	6/6
LCD-OLINUXINO-7TS Rev A	0 / 1	0 / 0 / 0 / 2	25	63	PASS	6/6
A13-LCD10TS Rev A	0 / 1	0 / 0 / 1 / 0	25	60	PASS	6/6
LCD-OLINUXINO-4.3TS Rev A	0 / 3	0 / 0 / 0 / 0	25	57	PASS	6/6
LS013B7DH03 TFT Module	0 / 0	0 / 0 / 0 / 6	24	52	PASS	6/6
RPi Speaker	0 / 0	0 / 0 / 0 / 0	30	63	PASS	6/6
BDVLS TI TXS0108E	0 / 16	0 / 0 / 0 / 2	31	59	PASS	6/6
MAX9709 Amplifier PCB	0 / 2	0 / 0 / 0 / 0	25	61	PASS	6/6
RPi ISM	0 / 3	0 / 0 / 0 / 0	31	67	PASS	6/6
RTL-SDR direct sampling diff amp	0 / 0	0 / 0 / 0 / 0	23	56	PASS	6/6
Telit SE880 GPS module for EFM32LG-STK3600	0 / 0	0 / 0 / 0 / 0	24	71	PASS	6/6

Native ERC/shorting per design were measured in run2 and are reported as corpus totals (367 / 347) in the chart above; the per-design native values for open nets and lost names come from the run3 baseline of the same files.

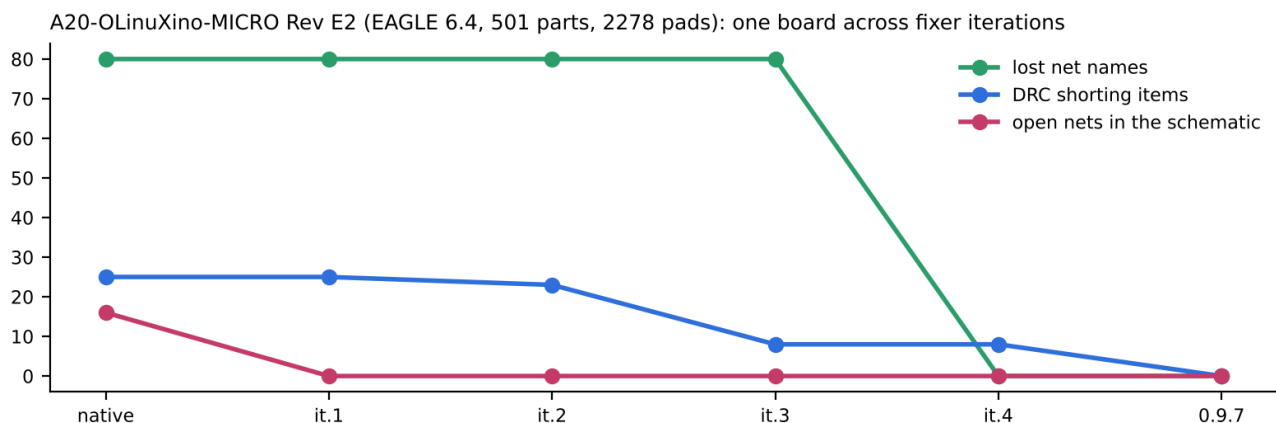
4. History: four full corpus runs



Run	When	Fixer	QC pass	ERC	Shorting	What changed / what was found
run1	7 Oct, morning	pre-release	14 / 18	-	-	First strict run. 3 self-check failures (iMX233 family: one injected fault was a no-op), 1 QC FAIL (Telit: a false pad-size mismatch, verifier bug), 1 crash (iMX233-MAXI: KiCad GetLength() assert on a circular outline). A10-Lime passed but hid a real defect: the board outline drawn inside a package left the copper pours almost unfilled (254 connections) - the QC gained an outline-length and a post-fill connectivity check for it.
run2	7 Oct, afternoon	0.9.0-0.9.5	18 / 18	13	65	Clean re-conversion with all run1 fixes. New: milling outside the board kept out of Edge.Cuts (MAXI: 29 invalid outline items), outline-validity check, printed solder-jumper crossings counted separately (BDVLS, LS013).
run3	7 Oct, night	0.9.6	18 / 18	13	77	New import path: board via kicad-cli (0.4-1.5 s) + schematic via GUI, import 60 s → 24-45 s per design. Shorting items rose 65 → 77 with the changed board import; investigated on A20: copper rectangles that bridge pads (board-mounted OR) were not treated as net ties.
run4	8 Oct, morning	0.9.7	18 / 18	16	26	Copper rectangles bridging two pads become net ties: A10/A20/A13 family 5/9/1/4/9 → 0, corpus 77 → 26. ERC 13 → 16: all power_pin_not_driven / pin_to_pin items, ±1 per board between runs on identical schematics (see §5).

Single-board view: the same board across fixer iterations

Before the corpus runs, every fixer version was measured on A20-OLinuXino-MICRO Rev E2. Iteration 1 (EAGLE "nc" pins wired in the source are made connectable) removed all 16 open nets, GND and VCC among them. The lost net names only came back in iteration 4, because every restored label had to be checked against the netlist: an earlier attempt put an SD_VCC label on the GND net. 0.9.7 removed the last 8 shorting items (OR bridges).



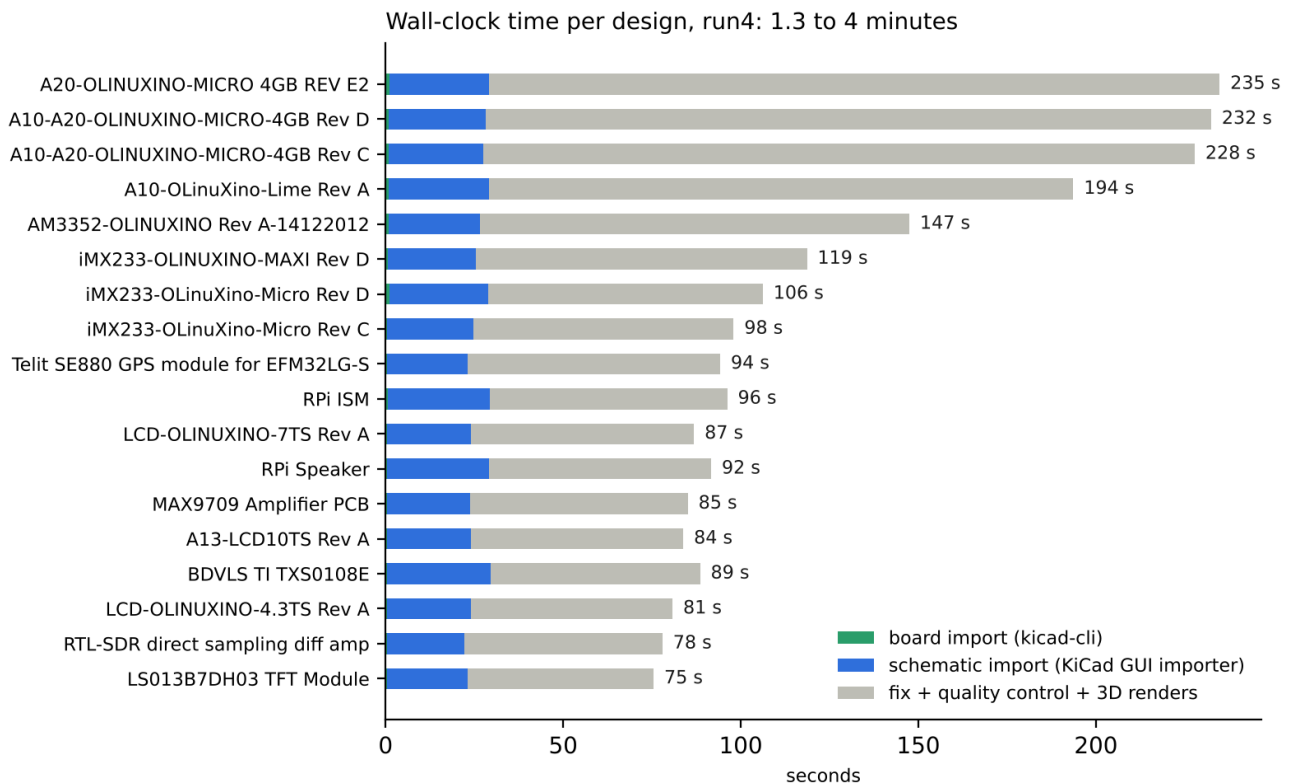
5. What remains after the fix

Design	ERC errors (type)	DRC shorting items	Reading
A10-A20-MICRO Rev C / Rev D, A20 Rev E2	2 / 3 / 2 (power_pin_not_driven, pin_to_pin)	0 / 0 / 0	power nets without an explicit source symbol; a PWR_FLAG decision for a human
A10-Lime Rev A	2 (pin_to_pin)	0	output-to-output pins in the source
A13-LCD10TS	1	0	
AM3352	2 (power_pin_not_driven, missing_power_pin)	1	
iMX233-MAXI / Micro C / Micro D	2 / 1 / 1	7 / 3 / 4	SMD solder-jumper pad pairs drawn as copper in the package
LCD-7TS	0	1	
BDVLS, LS013 (Z. Doczi)	0	2 / 6	printed solder jumpers: tracks run to the far pad of the jumper in the EAGLE source as well
other 7 designs	0	0	

Everything in this table is a property of the original EAGLE design that the conversion carries over rather than redesigning. Other DRC classes (clearance, solder-mask bridges, starved thermals, annular width) were not targeted and are listed in each project's report; the largest one, A20, violates its own 0.127 mm track-to-via clearance by 0.019 mm in 450+ places in EAGLE too.

Known weak spot. The ERC total moved 13 → 13 → 16 across runs whose schematic-side processing is identical; the differing items are all power_pin_not_driven. The power-flag placement is therefore not fully deterministic yet and is tracked as an open issue.

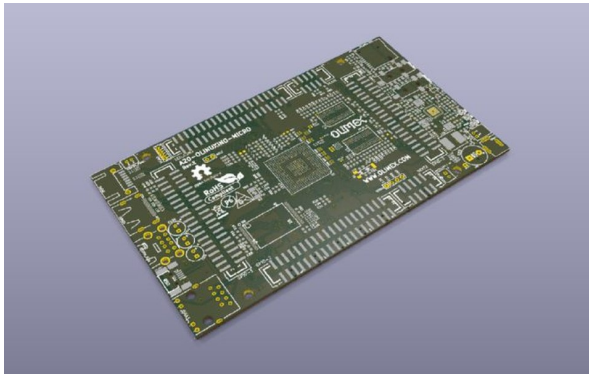
6. Timing



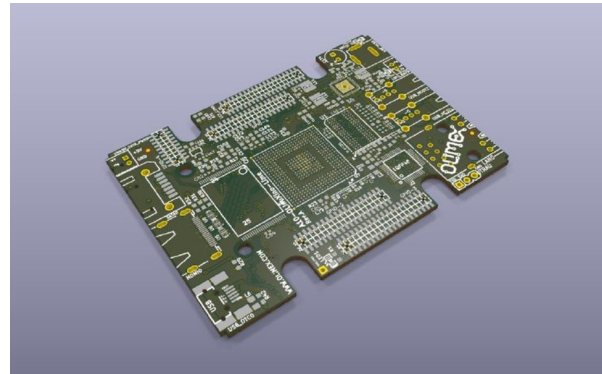
Import: 23-31 s per design, of which the board takes 0.4-1.1 s (kicad-cli) and the schematic 22-29 s (KiCad GUI importer, no CLI alternative). Fix + quality control + renders: 52-206 s. Of the latter, in run3 about 30% was 3D rendering and about 55% measurement (quality control, ERC/DRC passes, self-check); the fix steps themselves are a few seconds. No speed optimisation has been done on purpose: stability and verification first.

7. Converted boards

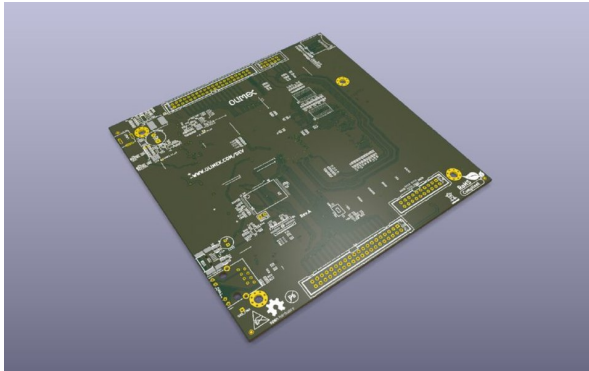
Rendered from the KiCad projects Eagle Exhumer produced (filled zones). EAGLE designs carry no 3D models, so the boards are bare. Olimex boards © Olimex Ltd (open hardware).



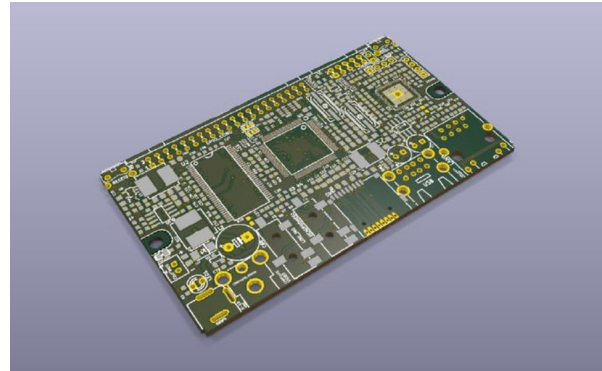
A20-OLinuXino-MICRO Rev E2 (Olimex)



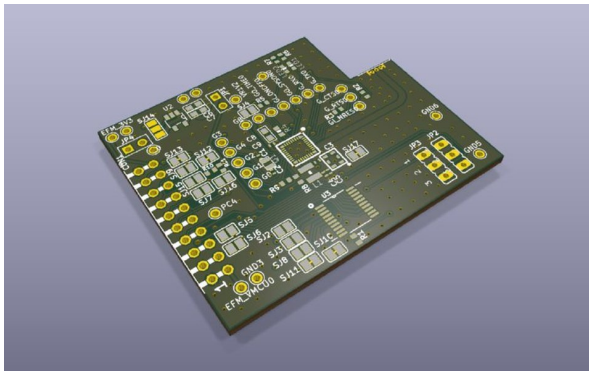
A10-OLinuXino-Lime Rev A (Olimex)



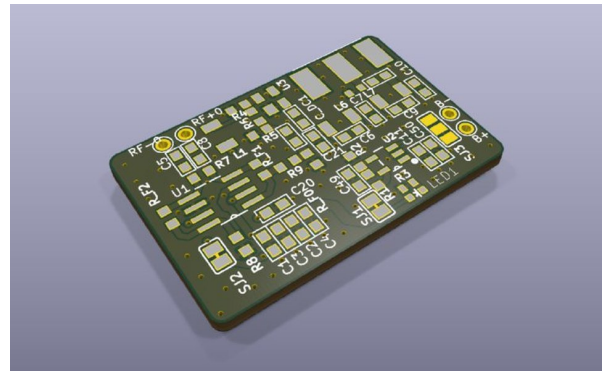
AM3352-OLinuXino Rev A (Olimex)



iMX233-OLinuXino-MAXI Rev D (Olimex)



Telit SE880 GPS module (Z. Doczi)



RTL-SDR direct-sampling differential amplifier (Z. Doczi)

8. Reproducing the numbers

Install the plugin (release zip via KiCad's Plugin and Content Manager), then from KiCad's Python: `python eagle2kicad_batch.py <folder with EAGLE .sch/.brd pairs>`. Every converted project contains `eaglefix_report.md` (verdict, findings, ERC/DRC summary), `eaglefix_metrics.json` (the numbers in this report, with stage timings) and `eaglefix_history.jsonl` (one line per run). The batch writes `batch_summary.md` per folder. Problems: the plugin's "Report a problem" button packs these files into a zip and opens a pre-filled GitHub issue; design files are included only on request.